

CSE232 – Computer Networks

Quiz 1 – Detailed Solutions

Section A

Topics Covered

- Non-persistent and persistent HTTP
- Round Trip Time (RTT)
- Transmission delay
- Propagation delay
- Web caching and cache hit rate
- Access-link utilization
- Bandwidth savings using web proxies
- End-to-end HTTP response time

Question 1

Compare Non-persistent HTTP and Persistent HTTP (without pipelining). Which one is more efficient for loading a webpage with 10 embedded images, and why?

Solution

The main difference between the two approaches is how the underlying TCP connection is used.

Non-persistent HTTP

In non-persistent HTTP, a separate TCP connection is established for *each* object that needs to be retrieved.

Suppose a webpage contains:

1 HTML file + 10 images

Then the total number of objects is

$$N = 1 + 10 = 11.$$

Therefore, non-persistent HTTP requires a separate connection for each of these 11 objects.

For each object, the client typically incurs connection-establishment overhead followed by the HTTP request/response exchange.

Thus, approximately,

$$\text{RTTs per object} = 2.$$

Hence,

$$\text{Total RTTs} = 11 \times 2 = 22 \text{ RTTs.}$$

Persistent HTTP without pipelining

In persistent HTTP, a single TCP connection remains open and can be reused for multiple objects. After the initial connection has been established and the base HTML file has been obtained, the same connection can be used to retrieve the embedded images.

Since pipelining is *not* being used, the requests are still handled sequentially. Nevertheless, there is no need to establish a new TCP connection for every image.

Thus, persistent HTTP avoids the repeated connection-establishment overhead associated with non-persistent HTTP.

Answer: Persistent HTTP is more efficient.

For a webpage containing 10 embedded images, persistent HTTP reuses a single TCP connection, whereas non-persistent HTTP establishes a new connection for each object. Therefore, persistent HTTP significantly reduces the RTT overhead and improves webpage loading efficiency.

Key idea:

Persistent HTTP $\Rightarrow$ Connection reuse $\Rightarrow$ Less RTT overhead $\Rightarrow$ Better efficiency

Question 2

A 100 KB file is sent over a 1 Mbps link. Calculate the transmission delay in milliseconds. Assume 1 KB = 1000 bytes.

Solution

The transmission delay is the amount of time required to push all the bits of a packet or file onto the communication link.

The formula is

$$d_{\text{trans}} = \frac{L}{R}$$

where

- L = size of the data in bits,
- R = transmission rate in bits per second.

Step 1: Convert the file size into bytes

The file size is

100 KB.

Using

1 KB = 1000 bytes,

we get

$$100 \text{ KB} = 100 \times 1000 = 100,000 \text{ bytes.}$$

Step 2: Convert bytes into bits

Since

$$1 \text{ byte} = 8 \text{ bits},$$

we have

$$L = 100,000 \times 8 = 800,000 \text{ bits}.$$

Step 3: Convert the link rate

The link rate is

$$R = 1 \text{ Mbps} = 1,000,000 \text{ bits/s}.$$

Therefore,

$$d_{\text{trans}} = \frac{800,000}{1,000,000} = 0.8 \text{ s}.$$

Step 4: Convert seconds into milliseconds

Since

$$1 \text{ s} = 1000 \text{ ms},$$

we get

$$0.8 \text{ s} = 0.8 \times 1000 = 800 \text{ ms}.$$

$$d_{\text{trans}} = 800 \text{ ms}$$

Therefore, the transmission delay is **800 milliseconds**.

Question 3

How many RTTs (Round Trip Times) are required in total to retrieve an HTML file and 4 images using Non-persistent HTTP? Exclude the initial DNS lookup and transmission times.

Solution

The webpage consists of:

$$1 \text{ HTML file} + 4 \text{ images}.$$

Therefore, the total number of objects is

$$N = 1 + 4 = 5.$$

With non-persistent HTTP, each object requires a new TCP connection.

For each object, two RTTs are required:

1. One RTT for establishing the TCP connection.
2. One RTT for sending the HTTP request and receiving the corresponding response.

Thus,

$$\text{RTTs per object} = 2.$$

For five objects,

$$\text{Total RTTs} = 5 \times 2.$$

Therefore,

$$\text{Total RTTs} = 10.$$

10 RTTs

Hence, excluding DNS lookup and transmission times, retrieving one HTML file and four images using non-persistent HTTP requires **10 RTTs**.

RTT Breakdown

Object	Connection RTT	Request/Response RTT
HTML	1	1
Image 1	1	1
Image 2	1	1
Image 3	1	1
Image 4	1	1
Total	5	5

Therefore,

$$5 + 5 = \boxed{10 \text{ RTTs}}.$$

Question 4

In Persistent HTTP (without pipelining), after the initial connection is established and the base HTML file is received, how many additional RTTs are needed to fetch 3 embedded objects?

Solution

The question explicitly states that:

- the TCP connection has already been established, and
- the base HTML file has already been received.

Therefore, we do *not* need to account for another TCP connection establishment.

For each embedded object, persistent HTTP without pipelining requires one request/response exchange.

Thus,

$$\text{RTTs per embedded object} = 1.$$

There are three embedded objects:

$$N = 3.$$

Hence,

$$\text{Additional RTTs} = 3 \times 1 = 3.$$

3 RTTs

Therefore, **3 additional RTTs** are needed to retrieve the three embedded objects.

Important Observation

The connection is already open. Hence, the TCP connection-establishment RTT is not repeated.

Persistent HTTP without pipelining $\Rightarrow$ 1 RTT per additional object

Question 5

A network has a cache hit rate of 40%. The access time to the local cache is 10 ms, while fetching from the origin server takes 100 ms. Calculate the average response time.

Solution

We have:

$$H = 40\% = 0.40$$

where H is the cache hit rate.

Therefore, the cache miss rate is

$$1 - H = 1 - 0.40 = 0.60.$$

The response time for a cache hit is

$$T_{\text{hit}} = 10 \text{ ms.}$$

The response time for a cache miss is

$$T_{\text{miss}} = 100 \text{ ms.}$$

The average response time is the weighted average of the two cases:

$$T_{\text{avg}} = HT_{\text{hit}} + (1 - H)T_{\text{miss}}.$$

Substituting the given values,

$$T_{\text{avg}} = (0.40)(10) + (0.60)(100).$$

Therefore,

$$T_{\text{avg}} = 4 + 60.$$

Hence,

$$T_{\text{avg}} = 64 \text{ ms.}$$

$$T_{\text{avg}} = 64 \text{ ms}$$

The average response time is therefore **64 milliseconds**.

Question 6

Assume a link has a propagation speed of 2×10^8 m/s and a length of 2,000 km. If a packet is 1,000 bits and the bandwidth is 1 Mbps, find the ratio of propagation delay to transmission delay.

Solution

We need to calculate two quantities:

1. Propagation delay
2. Transmission delay

and then calculate their ratio.

Step 1: Calculate propagation delay

The propagation-delay formula is

$$d_{\text{prop}} = \frac{d}{s},$$

where

- d = physical distance,
- s = propagation speed.

The distance is

$$2000 \text{ km.}$$

Converting kilometres to metres:

$$2000 \text{ km} = 2000 \times 1000 = 2,000,000 \text{ m.}$$

The propagation speed is

$$s = 2 \times 10^8 \text{ m/s.}$$

Therefore,

$$d_{\text{prop}} = \frac{2,000,000}{2 \times 10^8}.$$

Simplifying,

$$d_{\text{prop}} = 10^{-2} \text{ s.}$$

Thus,

$$d_{\text{prop}} = 0.01 \text{ s} = 10 \text{ ms.}$$

Step 2: Calculate transmission delay

The transmission-delay formula is

$$d_{\text{trans}} = \frac{L}{R}.$$

Here,

$$L = 1000 \text{ bits}$$

and

$$R = 1 \text{ Mbps} = 10^6 \text{ bits/s.}$$

Hence,

$$d_{\text{trans}} = \frac{1000}{10^6} = 10^{-3} \text{ s.}$$

Therefore,

$$d_{\text{trans}} = 1 \text{ ms.}$$

Step 3: Calculate the ratio

The required ratio is

$$\frac{d_{\text{prop}}}{d_{\text{trans}}} = \frac{10 \text{ ms}}{1 \text{ ms}}.$$

Therefore,

$$\frac{d_{\text{prop}}}{d_{\text{trans}}} = 10$$

or equivalently,

$$10 : 1.$$

$$\text{Propagation delay : Transmission delay} = 10 : 1$$

Interpretation: The propagation delay is ten times the transmission delay for the given link.

Question 7

A campus access link has a capacity of 10 Mbps. The average request size is 100,000 bits and the internal request rate is 120 requests/second. Calculate the minimum cache hit rate required to prevent the access link from being overloaded, i.e., to keep traffic below 10 Mbps.

Solution

We are given:

$$R_{\text{link}} = 10 \text{ Mbps}$$

$$L = 100,000 \text{ bits/request}$$

and

$$\lambda = 120 \text{ requests/s.}$$

The cache should reduce the traffic sent over the access link so that the external traffic does not exceed 10 Mbps.

Step 1: Calculate traffic without caching

The traffic generated by all requests is

$$T_{\text{raw}} = L \times \lambda.$$

Substituting,

$$T_{\text{raw}} = 100,000 \times 120.$$

Thus,

$$T_{\text{raw}} = 12,000,000 \text{ bits/s.}$$

Therefore,

$$T_{\text{raw}} = 12 \text{ Mbps.}$$

Without caching, the access link would have to carry

$$\boxed{12 \text{ Mbps}}$$

which is greater than its capacity of 10 Mbps.

Hence, caching is required.

Step 2: Determine the maximum fraction allowed on the link

Let H denote the cache hit rate.

The cache serves a fraction H of the requests locally.

Therefore, only the remaining fraction

$$1 - H$$

must cross the external access link.

Hence,

$$(1 - H)(12 \text{ Mbps}) \leq 10 \text{ Mbps.}$$

Step 3: Solve for the cache hit rate

Dividing by 12,

$$1 - H \leq \frac{10}{12}.$$

Therefore,

$$H \geq 1 - \frac{10}{12}.$$

Simplifying,

$$H \geq \frac{2}{12}.$$

Thus,

$$H \geq \frac{1}{6}.$$

In percentage form,

$$H \geq \frac{1}{6} \times 100 = 16.67\%.$$

$$H_{\min} = 16.67\%$$

Therefore, the minimum cache hit rate required is approximately **16.67%**.

Verification

At a 16.67% hit rate, approximately 83.33% of the traffic passes through the access link:

$$12 \times 0.8333 \approx 10 \text{ Mbps.}$$

Thus, the access link is brought down to approximately its 10 Mbps capacity.

Question 8

A company installs a web cache to save money on bandwidth. If their current WAN usage costs \$1000/month and the cache achieves a 60% hit rate, what is the new estimated monthly cost?

Solution

The original monthly WAN cost is

$$C_{\text{original}} = \$1000.$$

The cache hit rate is

$$H = 60\% = 0.60.$$

This means that 60% of the requests are served locally by the cache and therefore do not need to consume the corresponding WAN bandwidth.

Consequently, only

$$1 - H = 1 - 0.60 = 0.40$$

or 40% of the original WAN traffic remains.

Therefore, the new estimated cost is

$$C_{\text{new}} = (1 - H)C_{\text{original}}.$$

Substituting,

$$C_{\text{new}} = 0.40 \times \$1000.$$

Thus,

$$C_{\text{new}} = \$400.$$

\$400 per month

The estimated monthly WAN cost after introducing the cache is **\$400**.

Cost Saving

The monthly saving is

$$\$1000 - \$400 = \$600.$$

Thus, the cache provides an estimated monthly bandwidth-cost saving of

\$600.

Question 9

Consider a scenario where a popular news video (100 MB) is requested by 50 different students on a campus. Contrast the total bandwidth consumed on the university's external Internet link with and without a web proxy.

Solution

We are given:

$$\text{Video size} = 100 \text{ MB}$$

and

$$\text{Number of students} = 50.$$

We compare two scenarios.

Case 1: Without a Web Proxy

Without a proxy or web cache, every student's request must retrieve the video from the external Internet.

Therefore, the video must cross the university's external Internet link once for each student.

The total bandwidth consumed is

$$B_{\text{without}} = 50 \times 100 \text{ MB}.$$

Hence,

$$B_{\text{without}} = 5000 \text{ MB}.$$

Equivalently,

$$5000 \text{ MB} = 5 \text{ GB}$$

assuming

$$1 \text{ GB} = 1000 \text{ MB}.$$

Therefore,

$$B_{\text{without}} = 5000 \text{ MB} = 5 \text{ GB}$$

is consumed on the external link.

Case 2: With a Web Proxy

With a web proxy/cache, the first student's request causes the 100 MB video to be retrieved from the external Internet and stored in the proxy cache.

The remaining students can retrieve the same video from the local proxy.

Therefore, the external Internet link carries the video only once.

Thus,

$$B_{\text{with}} = 1 \times 100 \text{ MB} = 100 \text{ MB}.$$

Hence,

$$B_{\text{with}} = 100 \text{ MB}$$

Comparison

The bandwidth reduction is

$$5000 - 100 = 4900 \text{ MB}.$$

The reduction factor is

$$\frac{5000}{100} = 50.$$

Thus, the external-link bandwidth consumption is reduced by a factor of 50.

The percentage reduction is

$$\frac{5000 - 100}{5000} \times 100 = 98\%.$$

Without proxy = 5000 MB

With proxy = 100 MB

Bandwidth saving = 4900 MB

Percentage saving = 98%

Key idea: A web proxy avoids repeatedly transferring the same popular object across the external Internet link.

Question 10

A client uses Persistent HTTP (no pipelining) to fetch a 10 KB HTML file and two 50 KB images. The link rate is 1 Mbps and RTT is 100 ms. Calculate the total time from the first request until all data is received. Ignore handshake RTT and assume 1 KB = 8000 bits.

Solution

We are given:

HTML size = 10 KB

Image size = 50 KB

for each of two images.

The link rate is

$R = 1 \text{ Mbps.}$

The RTT is

$RTT = 100 \text{ ms.}$

The TCP handshake RTT is explicitly ignored.

The total time consists of:

Total time = RTT time + Transmission time

We calculate each part separately.

Step 1: Determine the number of RTTs

There are three objects:

1 HTML file + 2 images = 3 objects.

Because persistent HTTP is being used, the connection is reused.

However, pipelining is not being used. Therefore, the objects are requested sequentially.

The base HTML file requires one request/response RTT:

$$1 \text{ RTT.}$$

After the HTML file is received, each of the two images requires one additional RTT:

$$2 \text{ RTTs.}$$

Therefore,

$$\text{Total RTTs} = 1 + 2 = 3 \text{ RTTs.}$$

Since

$$RTT = 100 \text{ ms,}$$

the total RTT-related delay is

$$T_{\text{RTT}} = 3 \times 100 = 300 \text{ ms.}$$

Thus,

$$\boxed{T_{\text{RTT}} = 300 \text{ ms}}$$

Step 2: Calculate total amount of data

The HTML file is

$$10 \text{ KB.}$$

The two images together have size

$$2 \times 50 = 100 \text{ KB.}$$

Therefore, the total amount of data is

$$D_{\text{total}} = 10 + 100 = 110 \text{ KB.}$$

Hence,

$$\boxed{D_{\text{total}} = 110 \text{ KB}}$$

Step 3: Convert the total size into bits

The problem states that

$$1 \text{ KB} = 8000 \text{ bits.}$$

Therefore,

$$110 \text{ KB} = 110 \times 8000 = 880,000 \text{ bits.}$$

Thus,

$$D_{\text{total}} = 880,000 \text{ bits}$$

Step 4: Calculate transmission time

The transmission-delay formula is

$$d_{\text{trans}} = \frac{L}{R}.$$

Here,

$$L = 880,000 \text{ bits}$$

and

$$R = 1 \text{ Mbps} = 1,000,000 \text{ bits/s.}$$

Therefore,

$$d_{\text{trans}} = \frac{880,000}{1,000,000}.$$

Hence,

$$d_{\text{trans}} = 0.88 \text{ s.}$$

Converting to milliseconds:

$$0.88 \times 1000 = 880 \text{ ms.}$$

Thus,

$$T_{\text{trans}} = 880 \text{ ms}$$

Step 5: Calculate total time

The total time is

$$T_{\text{total}} = T_{\text{RTT}} + T_{\text{trans}}.$$

Substituting,

$$T_{\text{total}} = 300 \text{ ms} + 880 \text{ ms.}$$

Therefore,

$$T_{\text{total}} = 1180 \text{ ms.}$$

Converting to seconds,

$$1180 \text{ ms} = 1.18 \text{ s.}$$

$$T_{\text{total}} = 1180 \text{ ms} = 1.18 \text{ s}$$

Therefore, the total time from the first request until all data is received is **1.18 seconds**.

Important Formulas

$$d_{\text{trans}} = \frac{L}{R}$$

$$d_{\text{prop}} = \frac{d}{s}$$

$$T_{\text{avg}} = HT_{\text{hit}} + (1 - H)T_{\text{miss}}$$

$$T_{\text{external}} = (1 - H)T_{\text{raw}}$$

where L is the packet/file size in bits, R is the link rate, d is the physical distance, s is the propagation speed, and H is the cache hit rate.